Harrison He

Champaign, IL · Johannesburg, South Africa | (217) 202-6283 | harrisonhe.01150@gmail.com | harrisonhe.com | github.com/harrisonhe01150-hub

EDUCATION

University of Illinois Urbana-Champaign Expected May 2030

Bachelor of Science in Mathematics (transferring into Mathematics & Computer Science)
Coursework: MATH 241 Calculus III (honors section) · MATH 257 Linear Algebra with Computational Applications · CS 124 Intro to Computer Science I · MATH 231 credit by proficiency exam
Activities: GTO Illini (poker strategy) · Financial Engineering Club · Illinois Data Science Club

St John's College, Johannesburg Graduated 2025

Cambridge A-Levels: A Mathematics, A* Further Mathematics, A Physics, A Chinese, B Chemistry*

EXPERIENCE

Founder & Engineer — 店小力 Dianxiaoli, production AI sales agent for small merchants Apr 2026 – Present · Remote

*Incubated by **Du Xiaoman Financial** (Baidu's fintech arm), 小满摘星计划, since Sep 2026 · write-up: harrisonhe.com/dianxiaoli.html*

- Conceived, built and operate, alone, an autonomous sales agent inside WeChat Customer Service and WhatsApp: quotes from the merchant's catalogue, negotiates within owner-set floors, records orders, handles payment screenshots, escalates edge cases to the owner. **Live 24/7 for 3 paying merchants on two continents**; first deployment running since 2025.
- **Architecture:** one FastAPI service serving many merchants (multi-tenant since Sep 2026, each request routed to its store by the channel account it arrived on); deterministic Python around exactly one model call per turn — the model returns text plus tagged actions (`[ORDER:sku:qty]`, `[ESCALATE]`) that a deterministic executor validates and runs; exact-match catalogue retrieval by design; a rule-engine fallback takes over when a provider times out, so an outage degrades to scripted replies, never silence.
- **Testing and release: 21 suites / 1,458 checks** gate every release — scripted buyer scenarios with required and forbidden numbers, golden tests that run the previous release in a subprocess (`git archive`) and diff buyer replies word for word, mutation tests of the nightly audit, Playwright end-to-end runs against a live server; 23 versioned releases since May 2026; the gate has blocked two of my own. A tenant-migration bug that surfaced only in a process-restart E2E was written up as an RCA and turned into a repo rule.
- **Du Xiaoman incubation (Sep 2026 – present):** selected to build the standardised, multi-tenant edition for Du Xiaoman's small-business merchants (two assigned contributors, pilot access to its merchant base, joint company planned at scale); designed the consent-gated merchant-data interface for their risk models; delivered the multi-tenant core and industry packs three weeks ahead of schedule.
- **2026 AI Pioneer Future Talent Competition** (AI 先锋未来人才大赛, Feishu/ByteDance; 109 enterprise briefs): entered solo and **shortlisted for both the Du Xiaoman and Tec-Do (钛动) briefs**; built a coached deployment and demo for each (Jul–Aug 2026), from which the Du Xiaoman collaboration above followed.
- **How I work:** AI-assisted engineering — I run a two-agent spec → implement → review harness, own every spec, review every diff against the full gate and verify each release end-to-end in production myself; wrote the roadmap, partner API docs and an 18-page onboarding handbook.

Incoming Intern — Allan Gray, Johannesburg *(one of South Africa's largest independent asset managers; role and dates to be confirmed)* Dec 2026

PROJECTS

AI Study Tutor — handwritten past-paper grading 2025 – Present

- LLM pipeline (vision OCR → grading model → explanation and new-problem generation) marking handwritten answers by logical validity; FastAPI on Railway, Next.js on Vercel. **Hand-audit of a full paper found 9 of 15 AI marks were false negatives, all in one direction** — now a 13-case regression suite gating every change.

harrisonhe.com — *personal site built from scratch in HTML / CSS / JavaScript, including 青, a bring-your-own-key LLM chat*

Cambridge Research Project — regression analysis of cryptocurrency price series St John's College

SKILLS

Programming: Python (FastAPI, pandas; primary language), JavaScript / HTML / CSS (Next.js), Java (coursework), Git, Playwright, Linux; Railway (persistent volumes, environment-scoped secrets, egress allow-listing) and Vercel operations; WeChat and WhatsApp platform APIs.

AI & data engineering: LLM agent design with DeepSeek, Anthropic and OpenAI APIs — tagged-action output, deterministic executors, rule-engine fallback, background workers, multi-tenant routing, role-based field visibility; evaluation gates, golden and mutation tests, transcript audits; consent-gated data export. **Languages:** Mandarin and English, both fluent.

ACHIEVEMENTS

- **South African Mathematics Olympiad:** 4th (2021), 7th (2022), Top 50 (2023), Top 20 (2024, 2025) nationally; South Africa IMO selection training camp, Stellenbosch University, 2021–2024. **Wits Mathematics Competition:** Top 10, 2023–2025.
- South Africa Team Maths Competition: 2nd place 2021–2024, 4th 2025. South African Informatics Olympiad: first round 2023, 2024; Estonian Informatics Open: Top 15 within South Africa. Qualified Professional Level 1 Go player (China). 100+ hours volunteer olympiad tutoring, 2022–2025.